

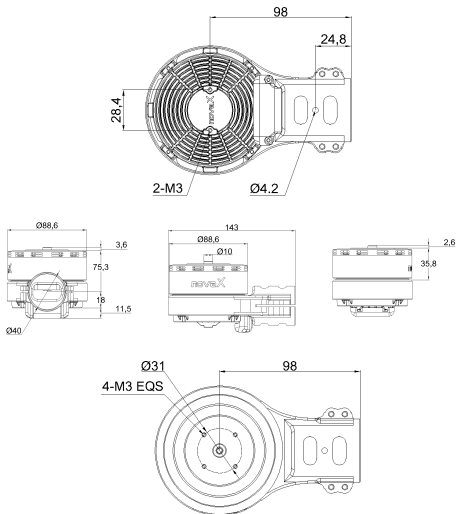
AM-8120

User Guide



AM-8120 Dimensions

• Dimensions (mm)



AM-8120 Specifications

• Basic Specifications

Stator Dimensions	$\Phi 90 \times 78.9 \text{ mm}$
Slots / Poles	36N40P (20 pole pairs)
Motor Weight	1150 g
KV Value	KV100
Rated Voltage	44.4 ~ 50.4 V
Max Current	54 A
Max Power	2700 W
Recommended Prop	28 ~ 30 inch (30×11)
Operating Temp	-20 ~ +85 °C

AM-8120 Performance Data

14S LiPo, 20%–100% throttle on static thrust stand. Measured values; rated specs may differ.

• Propeller 30 × 11

Throttle	Thrust (g)	Current (A)	Power (W)	Eff. (g/W)	Temp (°C)
20%	772	1.07	54	14.3	30
40%	3059	5.38	270	11.3	—
60%	6351	15.69	786	8.0	—
80%	10358	32.20	1605	6.5	—
100%	14708	54.27	2596	5.9	—

AM-8120 Installation

1. **Mounting** — Fix the motor to the arm through the bottom mounting holes. Follow the screw size and spacing on the drawing and use screws that do not reach the windings.
2. **ESC wiring** — Connect the three motor wires to the ESC outputs. To reverse the rotation, swap any two wires or reverse it in the ESC settings.
3. **Spin check** — Without a propeller, run at low throttle and check the direction and for abnormal noise.
4. **Propeller** — Fit a propeller matching the rotation and tighten the nut firmly.
Recommended: 28 ~ 30 inch (30×11).

AM-8120 Safety Notes

- Do not exceed the rated voltage. Rated: 44.4 ~ 50.4 V.
- Use an ESC rated with margin above the motor maximum current. Max current: 54 A.
- Always remove the propeller while wiring, configuring or bench testing.
- The motor can be hot right after operation; let it cool before touching.
- Keep metal debris away from the magnets; stop using the motor if the shaft is bent or the bearings are noisy.



<Detailed Manual>

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